

Sleep duration and musculoskeletal injury incidence in physically active men and women: A study of U.S. Army Special Operation Forces soldiers☆☆☆☆☆☆☆☆

Tyson Grier, MS^{a,*}, Eero Dinkeloo, MPH^a, Mark Reynolds, MD^a, Bruce H. Jones, MD, MPH^a

^a U.S. Army Public Health Center, Clinical Public Health and Epidemiology Directorate, 8977 Sibert Road, Aberdeen Proving Ground, MD, USA 21010



ARTICLE INFO

Article history:

Received 17 September 2019

Received in revised form 3 January 2020

Accepted 5 January 2020

Keywords:

Physical Training

Exercise

Age

Military

ABSTRACT

Purpose: In physically demanding occupations or athletic populations, sleep has been shown to be an important factor for both performance and recovery. However, evidence is limited on sleep duration and its association with musculoskeletal injury risk in physically active populations. Therefore, the purpose of this analysis was to evaluate the relationship between sleep duration and musculoskeletal injury incidence in a population of physically active men and women.

Methods: Data were collected via electronic survey and analyzed for 7,576 soldiers from the United States Army Special Operations Command.

Results: This population was mostly men (95%) ≤ 35 years old (70%). The incidence of musculoskeletal injury was 53% over the twelve months before survey administration. After adjusting for other covariates, soldiers who slept ≤ 4 hours were 2.35 (95% CI: 1.89–2.93, $p < 0.01$) times more likely to experience a musculoskeletal injury compared with those who slept eight ≥ 8 hours. When stratified by age (≤ 35 years, > 35 years), older men were found to have significantly higher risk of a musculoskeletal injury than their younger counterparts across all sleep duration groups.

Conclusions: These findings suggest that sleep duration may be inversely related to risk of musculoskeletal injury. It is possible that interventions targeted at sleep may have a positive impact on prevention of musculoskeletal injuries within physically active populations.

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Introduction

Sleep deficiency is defined by the National Institutes of Health as a lack of sleep, poor quality sleep, or sleeping at the wrong time of day¹. Ongoing sleep deficiency has been associated with impaired reaction times, memory, alertness, and a higher risk of many chronic conditions, including obesity, diabetes, hypertension, and heart disease^{2–6}. Sleep deficiency may also have an effect on immune function, exhibited in one study showing a reduced resistance to the common cold after weeks of shorter sleep duration⁷. From an athletic performance

standpoint, sleep deprivation has been negatively associated with both endurance and anaerobic performance⁸. In addition, poor sleep quality has been linked to a lower likelihood of meeting aerobic and resistance training recommendations as recommended by the Centers for Disease Control and Prevention and the American College of Sports Medicine⁹. Anecdotally, Soldiers in Basic Combat Training have reported feeling “weak” or “slow” as a result of sleep deficiency¹⁰.

Sleep deficiencies in the workplace have been shown to place workers at a higher risk of injury. Injuries occurring in the workplace have been defined as a discrete occurrence or accident leading to physical or mental harm¹¹. Previous studies investigating sleep deficiencies and work related injury risk¹¹ indicated increasing work-related injury risk with decreasing sleep hours and associations between sleep problems and work-related injuries^{11,12}. By contrast, only a limited number of studies have attempted to explore the effect of sleep duration on musculoskeletal injuries which are typically not captured from workplace injury studies. Two studies that have investigated sleep duration and musculoskeletal injuries were in adolescent athletes. One study indicated an increased odds of 1.7 for a sports-related injury when comparing an average night's sleep of < 8

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☆☆ The views expressed in this article are those of the author(s) and do not necessarily reflect the official policy of the Department of Defense, Department of the Army, U.S. Army Medical Department or the U.S.

☆☆☆ Conflict of Interest: The authors declare that there are no conflicts of interest.

☆☆☆☆ Funding: None.

* Corresponding author at: Tyson Grier, U.S. Army Public Health Center, Clinical Public Health and Epidemiology Directorate, 8977 Sibert Road, Aberdeen Proving Ground, MD, USA 21010.

E-mail address: Tyson.L.Grier.Civ@mail.mil (T. Grier).

hours to ≥ 8 hours¹³, whereas another demonstrated a correlation between acute short sleep duration (< 6 hours the night before) and a sports-related injury¹⁴. In the army, approximately two-thirds of injuries are musculoskeletal overuse injuries which are primarily attributed to physical training or repetitive-related activity¹⁵. With an estimated 42% of military personnel sleeping < 5 hours a night¹⁶, there is an opportunity to target and mitigate the effects of this potential modifiable risk factor on musculoskeletal injuries. The Performance Triad Guide which discusses the importance of sleep, activity, and nutrition in the U.S. Army soldiers, states that Soldiers require 7–8 hours of sleep every 24 hours to maintain their physical and psychological health¹⁷. The National Sleep Foundation also recommends that adults 18–64 years old get 7–9 hours of sleep. For sleep is essential for well-being, health, and performance. The purpose of this analysis was to investigate the relationship between sleep duration and musculoskeletal injury incidence in U.S. Army Special Operations Forces Soldiers.

Methods

Participants

Participants were men and women in the U.S. Army Special Operations Command (USASOC). The eligible population consisted of Special Operations Forces (SOF) Soldiers. This includes men with an 18 series military occupational specialty (MOS) (Special Forces), men and women with a 38 MOS series (Civil Affairs), men and women with a 37 MOS series (Psychological Operations) and men and women in active duty Special Operations Forces Support Staff positions. Men with an MOS series of 18 and men and women with a MOS series of 37 and 38 will be referred to as SOF operators. Men and women in support staff positions will be referred to as SOF support. Both groups combined (SOF operators and SOF support staff) will be referred to as SOF soldiers. Participants were excluded from this analysis if their survey did not contain valid responses to sex or sleep questions. This evaluation was reviewed and approved by the U.S. Army Public Health Center Review Board.

Survey

This is a secondary analysis of the Preservation of the Force and Family (POTFF) Wave IV Needs Assessment Survey. This survey was administered electronically to the United States Special Operations Command (USSOCOM) from 29 March to 9 May 2016. The primary goal of the POTFF survey was to provide commanders with a clearer understanding of the needs and wellness status among USSOCOM Service Members and their families. The survey consists of 95 items crossing a wide spectrum of demographic (sex, age, rank...), behavioral (sleep duration, tobacco use...), activity (physical training...), and psychological domains.

Exposure and outcome

The exposure of interest in this analysis was hours of self-reported sleep per night. The question on the survey asked “How many hours of actual sleep did you get at night (This may be different than the number of hours you spend in bed).” The outcome of interest in this analysis was the incidence of “musculoskeletal injury”. The question on the survey asked “How many musculoskeletal injuries have you sustained in the past 12 months”.

Data analysis

The Statistical Package for the Social Sciences (SPSS®), Version 21.0, was used for all statistical analysis. Frequencies, means, and

standard deviations were calculated for demographic characteristics and physical activity. Age groups were categorized as ≤ 35 and > 35 years old as the number of older SOF soldiers represented $< 30\%$ of the population when using higher 5-year cut-points. The SOF soldiers were also categorized into SOF operators and SOF support because of previous findings suggesting differences in physical performance and the amount of time spent performing physical fitness.¹⁸ Continuous variables were categorized into quartiles excluding sleep duration. Sleep duration was categorized by the number of hours slept from ≤ 4 hours of sleep per night to ≥ 8 hours of sleep per night. Hours of sleep per night, reported in hours and minutes of sleep, were rounded up to the nearest hour if the number of minutes was > 30 . An ANOVA was used to determine differences in the number of hours slept and physical training time. A chi-square was used to identify linear trends and to determine any differences in the number of hours slept using personal and occupational characteristics. Injury incidence was calculated as the number of service members that reported experiencing one or more musculoskeletal injuries (as reported on the survey) in the twelve months before survey administration, divided by the total number of service members surveyed. In a previous investigation, self-reported injuries were shown to be consistent with medical record data¹⁹. Injury incidence was also stratified by age and the number of hours slept. A post hoc power analyses was performed for men and women after analyzing musculoskeletal injury risk stratified by age group and the number of hours slept given that the individual cell sizes for women were small. Musculoskeletal injury risk ratios (RR) and 95% confidence intervals (95% CI) were calculated for age and the number of hours slept. Risk factors for musculoskeletal injuries were calculated using a univariate logistic regression, reported as odds ratios (OR) with 95% CI. Potential risk factors for a musculoskeletal injury, as identified by the univariate regression ($p < 0.05$), were explored using a multivariate logistic regression. The final multivariate logistic regression reported adjusted odds ratios (AORs) with 95% CI.

Results

Descriptive statistics

Of the 8,553 active duty soldiers who completed the POTFF survey, 1,033 were excluded for missing data. The final study population included 7,576 soldiers, ranging in age from 18 to 60 years old. Most of the SOF soldiers (70%) were aged 18–35 years old. Overall injury incidence for the twelve months before survey administration was 53% (men 53%; women 52%). Table 1 displays personal and occupational characteristics, as well as physical training, grouped by the number of hours slept per night. Most participants slept 6–7 hours per night, and this same trend is observed in both men and women independently. However, a higher percentage of women slept ≥ 8 hours per night compared with the population of men. A higher percentage of those who were younger, SOF operators, and enlisted slept ≥ 8 hours per night compared with those who were older, SOF support staff, and officers. A linear trend was shown for those using nicotine products with a decrease in nicotine usage as the number of hours slept per night increased ($p < 0.01$). Unlike the other variables (sex, age, job classification, and rank), the percentage of nicotine use was calculated by column to illustrate the linear trend. For physical training, there were no differences in the amount of vigorous physical activity performed and the number of hours slept per night. However, those who slept ≤ 4 hours a night performed a greater amount of moderate physical activity, and those who slept ≥ 8 hours performed a greater amount of strength training.

Figure 1 displays musculoskeletal injury incidence by the number of hours slept per night. Men and women had generally similar musculoskeletal injury incidence when stratified by hours of sleep per

Table 1
Personal characteristics, occupational characteristics, and physical training by sleep duration

Variable	n (%)	≤ 4 hours of sleep n (%)	5 hours of sleep n (%)	6 hours of sleep n (%)	7 hours of sleep n (%)	≥ 8 hours of sleep n (%)	p-value
Sex							<0.01*
Men	7,202 (95)	650 (9)	1,386 (19)	2,122 (30)	1,941 (27)	1,103 (15)	
Women	374 (5)	37 (10)	57 (15)	101 (27)	98 (26)	81 (22)	
Age							<0.01*
≤ 35 years	5,305 (70)	440 (8)	947 (18)	1,517 (29)	1,465 (28)	936 (18)	
> 35 years	2,264 (30)	247 (11)	494 (22)	703 (31)	573 (25)	247 (11)	
Job classification							<0.01*
SOF operator	4,236 (57)	352 (8)	771 (18)	1,218 (29)	1,174 (28)	721 (17)	
SOF support	3,214 (43)	324 (10)	646 (20)	970 (30)	829 (26)	445 (14)	
Rank							<0.01*
Enlisted	5,763 (76)	571 (10)	1,138 (20)	1,652 (29)	1,462 (25)	940 (16)	
Warrant officer	425 (6)	34 (8)	87 (21)	122 (29)	134 (32)	48 (11)	
Officer	1,362 (18)	79 (6)	213 (16)	443 (33)	435 (32)	192 (14)	
Nicotine products***							<0.01*
Yes	2,262 (30)	244 (36)	531 (37)	687 (31)	525 (26)	275 (24)	
No	5,250 (70)	440 (64)	903 (63)	1,516 (69)	1,497 (74)	894 (76)	
Average amount of physical activity per week in minutes ± SD							
Vigorous exercise	6,933	609 177±127	1,307 174±118	2,046 171±123	1,904 173±119	1,067 177±125	0.68
Moderate exercise	6,315	567 178±166	1,202 159±157	1,874 154±153	1,715 156±150	957 162±146	0.02**
Strength training	6,899	609 225±144	1,301 223±148	2,031 214±137	1,891 215±137	1,067 228±136	0.03**

Italic values indicate the average number of minutes of physical activity per week.

* Chi-square.

** ANOVA.

*** Chi square for trend $p < 0.05$ / SD = standard deviation.

night. However, there was a significant difference in musculoskeletal injury incidence between those strata, as well as a linear trend for musculoskeletal injury incidence to decrease with increasing hours of sleep per night ($p < 0.01$).

Table 2 displays univariate (unadjusted) and multivariate (adjusted) associations of various risk factors with musculoskeletal injury risk odds ratios. For the multivariate model, age was chosen over rank because age and rank are typically correlated (most of the enlisted soldiers were younger and the officers were older). The unadjusted odds ratios indicated that older age, officers, less sleep, increased vigorous exercise, and a moderate amount of

moderate exercise were risk factors for injury. When added to the multivariate model, older age, less sleep, increased amounts of vigorous exercise, and moderate amounts of moderate exercise were identified as risk factors for injury. After adjustment, older age (AOR=1.62) and less sleep (AOR=1.24–2.35) were the strongest risk factors for injury.

Because age and fewer hours of sleep per night were the strongest risk factors for injury, injury risk ratios were calculated to determine differences between age and hours of sleep per night for men and women. However, when dividing age groups into ≤ 35 and > 35 years old, the cell size for women became relatively small. As a result

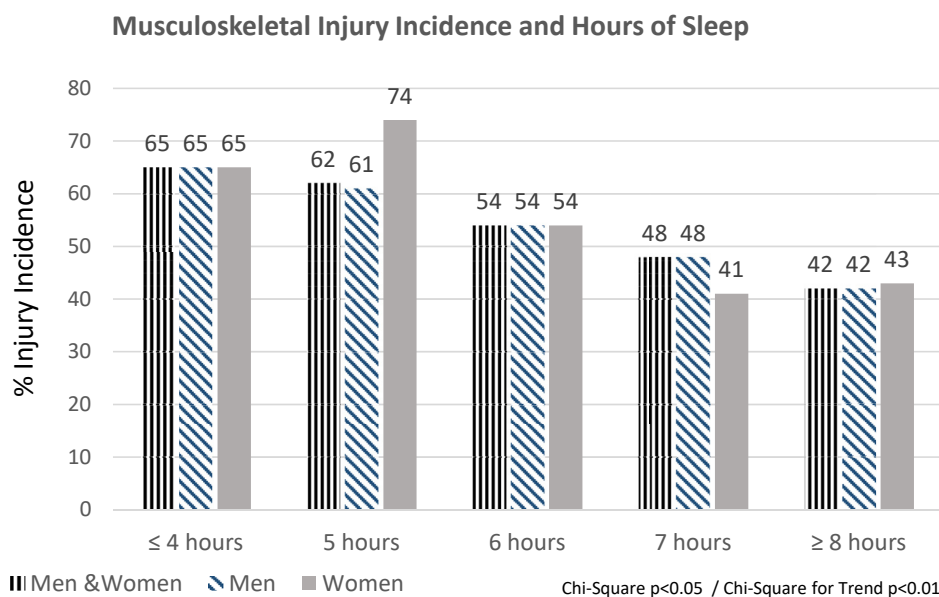


Figure 1. Musculoskeletal injury incidence and hours of sleep per night. There was a linear trend for musculoskeletal injury incidence to decrease with increasing hours of sleep per night ($p < 0.01$).

Table 2

Unadjusted and adjusted odds ratios for personal characteristics, occupational characteristics, sleep duration, and physical training associated with musculoskeletal injury risk

Category	Variable	n	Reported injury (%)	Unadjusted odds ratio (95% CI)	p-value	n	Adjusted odds ratio (95% CI)	p-value
Sex	Men	7,202	53	1.00		5,901	1.00	
	Women	374	52	0.97 (0.79–1.20)	0.78	314	1.04 (0.83–1.31)	0.72
Age	≤ 35 years	5,305	49	1.00		4,387	1.00	
	> 35 years	2,264	62	1.68 (1.52–1.86)	<0.01	1,828	1.62 (1.44–1.81)	<0.01
Job classification	Operator	4,236	54	1.0				
	Support	3,214	52	0.93 (0.85–1.02)	0.14			
Rank	Enlisted	5,763	52	1.00				
	Warrant officer	425	54	1.07 (0.88–1.30)	0.52			
	Officer	1,362	56	1.16 (1.03–1.31)	0.01			
Nicotine products	No	5,250	53	1.00				
	Yes	2,262	54	1.05 (0.95–1.16)	0.36			
Sleep***	≤ 4 hours	687	65	2.56 (2.11–3.11)	<0.01	553	2.35 (1.89–2.93)	<0.01
	5 hours	1,443	62	2.24 (1.91–2.62)	<0.01	1,176	2.06 (1.73–2.46)	<0.01
	6 hours	2,223	54	1.64 (1.42–1.89)	<0.01	1,846	1.53 (1.30–1.79)	<0.01
	7 hours	2,039	48	1.30 (1.12–1.50)	<0.01	1,698	1.24 (1.05–1.45)	0.01
Vigorous exercise***	≥ 8 hours	1,184	42	1.00		942	1.00	
	≤ 90 min	2,294	51	1.00		2,077	1.00	
	91–135 min	1,407	52	1.04 (0.91–1.19)	0.59	1,278	1.02 (0.88–1.17)	0.82
	136–225 min	1,981	53	1.06 (0.94–1.19)	0.38	1,764	1.03 (0.90–1.17)	0.67
Moderate exercise	≥ 226 min	1,251	55	1.16 (1.01–1.33)	0.04	1,096	1.24 (1.06–1.45)	<0.01
	≤ 45 min	1,665	51	1.00		1,641	1.00	
	46–105 min	1,625	54	1.17 (1.02–1.34)	0.03	1,607	1.17 (1.02–1.35)	0.03
	106–225 min	1,902	55	1.20 (1.05–1.36)	<0.01	1,862	1.17 (1.02–1.34)	0.03
Strength training	≥ 226 min	1,123	52	1.08 (0.92–1.25)	0.35	1,105	1.05 (0.89–1.23)	0.59
	≤ 135 min	2,690	53	1.05 (0.93–1.20)	0.43			
	136–180 min	826	55	1.13 (0.95–1.34)	0.17			
	181–300 min	1,904	53	1.04 (0.91–1.19)	0.57			
	≥ 301 min	1,479	52	1.00				

*** Chi-square for trend $p < 0.05$.**Table 3**

Musculoskeletal injury risk by age and sleep duration for men

Age	≤ 4 hours of sleep			5 hours of sleep			6 hours of sleep			7 hours of sleep			≥ 8 hours of sleep			RR 95% CI* (≤ 4 hours of sleep / ≥ 8 hours of sleep)
	Injured			Injured			Injured			Injured			Injured			
	No	Yes	Injury (%)	No	Yes	Injury (%)	No	Yes	Injury (%)	No	Yes	Injury (%)	No	Yes	Injury (%)	
≤ 35 years***	167	245	60	392	512	57	702	733	49	757	626	45	538	333	38	1.56 (1.39-1.75)**
> 35 years***	63	175	74	148	332	69	242	412	60	243	314	56	107	125	54	1.37 (1.19-1.57)**
RR 95% CI* (>35/≤ 35)	1.23 (1.10-1.38)**			1.22 (1.12-1.33)**			1.23 (1.14-1.33)**			1.25 (1.14-1.37)**			1.41 (1.22-1.63)**			

Bold values indicate the percent injured.

* Risk ratio and 95% confidence interval.

** $p < 0.05$.*** Chi-square for trend $p < 0.05$.

of these small cell sizes, a post hoc power analysis was performed. The analysis for women indicated that the power of these smaller cell sizes was < 0.20 , thereby lacking the power to detect differences in musculoskeletal injury risk stratified by age group and the number of hours slept. Owing to being underpowered, the results for women are not included. However, the power analysis performed for men indicated adequate power (> 0.80) to detect differences in musculoskeletal injury risk stratified by age group and the number of hours slept (Table 3). The analysis revealed that older men had a higher risk of a musculoskeletal injury when compared with younger men sleeping the same number of hours per night. It was also indicated that younger and older men who slept ≤ 4 hours per night were 56% and 37% more likely to experience a musculoskeletal injury than those sleeping 8 or more hours per night, respectively. When making the most extreme comparison, older men (> 35 years) who slept ≤ 4 hours per night versus men who were younger (≤ 35 years) and slept for ≥ 8 hour per night were 92% more likely to experience a musculoskeletal injury (RR 95% CI [> 35 years ≤ 4 hours of sleep / ≤ 35 years ≥ 8 hours of sleep] 1.92 [1.72–2.16] $p < 0.01$). There was also a linear trend for younger and older men showing that as the hours of sleep increased, the risk of a musculoskeletal injury decreased.

Discussion

This secondary analysis investigated sleep duration and musculoskeletal injury incidence in SOF soldiers. A primary finding of this analysis was that as sleep duration decreased, risk of a musculoskeletal injury incidence increased; this outcome has not been shown in an active adult population to date. Age was also shown as a primary risk factor with older men having a higher risk of experiencing a musculoskeletal injury. When further explored by sleep duration group, older men had an even higher risk of a musculoskeletal injury compared with younger men and exhibited the same overall trend of increased musculoskeletal injury risk with decreasing sleep duration.

The overall injury rate in this analysis was consistent with other studies of military personnel^{18,20,21}. As mentioned earlier, in the army, approximately two-thirds of injuries are musculoskeletal overuse injuries which are primarily attributed to physical training, overtraining, or repetitive-related activity¹⁵. In a previous investigation of musculoskeletal injuries among 743,000 military personnel, 82% of all musculoskeletal injuries were classified as overuse injuries (inflammation and pain)²². This tends to be different from studies investigating workplace injuries which are usually not musculoskeletal overuse injuries but rather accidental injuries or injuries as a result

of an acute exposure to external forces or substances. In studies performed in the workplace, an increased risk of a workplace injury was associated with decreasing sleep duration^{11,23}. In a systematic review of sleep problems and work injuries, 13% of work injuries could be attributed to sleep problems¹¹. These findings are related to the present study where musculoskeletal injury risk also increased with decreasing sleep duration.

As sleep duration decreased, musculoskeletal injury increased. Previous literature regarding the association of sleep duration and musculoskeletal injury risk is limited. Nevertheless, there are a few studies that have explored this relationship. In a study of adolescent athletes (15 years old $\pm$ 1.5 years) those who slept $\leq$ 8 hours per night were 70% more likely to experience an injury (injuries were obtained from the school's athletic department) than those who slept $\geq$ 8 hours per night¹³. By contrast, no interaction between sleep duration and injury was found among elite Australian footballers²⁴, although this investigation was limited in size (22 players). In another study of adolescent athletes, it was shown that increases in training loads and intensities were associated with decreasing sleep volume and resulted in a 2.25 times higher risk of injury compared to those with no changes in sleep duration and physical training²⁵. It is suggested that a sudden change in the balance of training load and rest may result in increased stress on the musculoskeletal system, thereby causing fatigue and functional impairment contributing to injury risk²⁵. Similar to athletes, soldiers also have variable physical training loads which may increase or decrease dependent on deployment schedules and current mission requirements. If their current sleep duration is inadequate to maintain these augmented physical training loads or heavier workloads, then they may be at a higher risk of experiencing a musculoskeletal injury.

Musculoskeletal injury risk was slightly higher among those performing moderate exercise and highest among those who performed the greatest amounts of vigorous exercise. In a study of elite team sport athletes, greater amounts of very high velocity running were associated with a greater risk of a lower body soft tissue injury. However, they found that greater distances covered by both mild and moderate running acceleration intensities offered a protective effect against soft tissue injury.²⁶ In another study investigating high velocity running (sprinting) and frequency, no difference in injury risk was observed²⁷. The differences in these findings compared with the current analysis could be attributed to how vigorous exercise was defined. In the POTFF survey, vigorous training was defined as exercise that causes heavy sweating or a large increase in breathing or heart rate. So vigorous training is referring to multiple types of vigorous exercise activities and not just high velocity running or sprinting. In regard to sleep, the effect of sleep deficiency on physical training could lead to earlier or higher levels of fatigue. Higher levels of fatigue have been associated with an increased risk of injury. Studies regarding fatigue have shown decrements in proprioceptive ability,²⁸ changes in gait²⁹, balance,³⁰ ligament laxity³¹, and alterations in muscle activity²⁸.

Older men were at a higher risk of experiencing a musculoskeletal injury. When further explored, musculoskeletal injury risk increased as the number of hours of sleep decreased for both younger and older men, with older men having an even greater risk of a musculoskeletal injury. Literature is limited as it relates to sleep and its effects on musculoskeletal injury and age. Most studies to date have focused exclusively on adolescent athletes. The results from this study suggest that both sleep and age may be important factors when considering injury risk. On its own, older age is a commonly cited risk factor for injury^{32–34}. The findings from this analysis reaffirm that conclusion; as older age (>35 years) was shown to increase musculoskeletal injury risk by 62%. However, when comparing age across sleep duration groups, it is clear that the presence of both of these factors act to elevate risk of a musculoskeletal injury to even higher levels for men.

While age is not a modifiable risk factor, this may provide direction for higher risk groups to target with sleep related interventions.

A potential limitation of this study was self-reported and not actually measured data. However, previous studies investigating the validation of self-reported compared with actual collected data have shown high correlations^{19,35,36}. Another potential limitation was the underrepresentation of women when stratified by age and the number of hours slept. In the future, a greater number of women would be needed to determine any differences between injury risk, age, and hours slept. This analysis was also performed on SOF soldiers who may differ from the normal population of generally healthy adults but have similarities to first responders. When comparing SOF soldiers with a normal population of generally healthy adults, there may be some broad differences in physical fitness and the ability to perform specialized training. SOF soldiers may differ from the general population in that they have to meet physical fitness training requirements on a bi-annual basis. In meeting these requirements, soldiers generally perform regular physical training with their units and during their own personal time. They are also required to meet and train for multiple operational and specialty requirements to be mission ready. However, like the general population, it is also recommended that soldiers get 7 or more hours of sleep to maintain their physical and psychological health.

Conclusion

It is recommended that adults 18–64 years old get 7 or more hours of sleep per night³⁷. In this evaluation, only 43% of the SOF soldiers reported sleeping 7 or more hours. This is not uncommon in military populations, where sleep disturbance is a widely reported issue^{9,16,38,39}. Results from this study suggest that even after adjustment for sex, age, and amount of physical activity, sleep duration shorter than 8 hours may be associated with an increased risk of a musculoskeletal injury. In addition, men over the age of 35 years had an even greater risk of experiencing a musculoskeletal injury than younger SOF soldiers in the same sleep durations groups. With most soldiers sleeping less than 7 hours per night, interventions to improve sleep duration may have a positive impact on musculoskeletal injury prevention within physically active populations.

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